

M Balaraman

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EDUCATION

Chennai Institute of Technology - Chennai , India

B.Tech. Information Technology , Pre-Final year

2023 – 2027

Current CGPA: 8.45/10.0

Coursework: Computer Architecture, Computer Networks, Operating Systems, Database Management Systems

EXPERIENCE

Software Development Engineer Intern, Amazon – Hyderabad, India

July 2026 - Present

- Built a **metrics dashboard** for **Supply Chain Shipment**, giving stakeholders real-time visibility into shipment health.
- Contributed to an **automation project** streamlining onboarding of new tenants to the **Shipment Tracking** service.
- Automated infrastructure and deployments with **AWS CDK** and **Pipelines**, enabling safe, repeatable releases through CI/CD.

Software Engineering Intern, FERN Galaxy, Microsoft – Bangalore, India

May - July 2026

- Worked in the **Graph Connectors Copilot Team**, building MCP-based solutions for connector onboarding and management.
- Developed an **MCP Server** in the core repository for LLM-driven automation of connector creation and third-party configuration.
- Enhanced the Admin Portal with **AI agents and automation**, reducing connector setup and publishing effort by **over 50%**.

Software Engineering Intern , Krown – Chennai ,India

Jan - Mar 2026

- Built scalable features for a **React Native** app and **Next.js** web platform using **WebSockets** for real-time updates and analytics.
- Developed secure, scalable **Razorpay** payment and booking flows with Node.js webhooks for transactions and confirmations.
- Improved security and performance using multi-device authentication, **httpOnly** cookies, and optimized **PostgreSQL** queries.

PROJECTS

MentorConnect | Next.js, React, .NET Core, Microservices, gRPC, RabbitMQ, Docker, SignalR

Link | Apr 2026

- Architected **.NET Core microservices** backend using **Docker** and **design patterns**, communicating via high-performance **gRPC**.
- Built three distinct **Next.js portals** (Admin, Mentor, Mentee) secured by **JWT** and **role-based access** to manage user workflows.
- Delivered seamless real-time **chat** and notifications using **SignalR** within a containerized **RabbitMQ** event-driven infrastructure.

AI Document Assistant (RAG) | React, TypeScript, Tailwind CSS, Python, LangChain, Supabase.

Link | Apr 2026

- Built the Python backend with **Retrieval-Augmented Generation (RAG)** for fast document processing and semantic search.
- Built a highly responsive React frontend using Tailwind CSS, delivering a **seamless interactive chat** for document analysis.
- Created a **secure full-stack application** using Supabase for reliable, persistent user authentication and real-time chat storage.

Drone Automation | Python , DroneKit.,PyMavlink ,GCS.

Link | Mar 2025

- Established a stable communication link between the drone and ensuring real-time telemetry and control signals.
- Built an autonomous drone navigation system with **DroneKit-Python** to execute GPS-based waypoint with minimal user input.
- Integrated a **Pi Camera and YOLOv5** to capture and analyze QR codes during landing, ensuring delivery zone detection..

TECHNICAL SKILLS

C	TypeScript	Git	Node.js	SQL
C++	React.js	Github	SpringBoot	DesignPattern
Python	Next.js	JavaScript	DSA	MongoDB
Java	Express.js	Linux	OOPS	Micronaut
Go	Fast API	KeyCloak	Docker	Kotlin

CERTIFICATIONS & COURSES

AWS - AWS Certified Cloud Practitioner	Link	2024
JLPT - Japanese Language Proficiency Test (N4)	Link	2025
NPTEL - Database Management System	Link	2023
NPTEL - Problem Solving using C	Link	2023
CISCO - Networking Essentials	Link	2023

COMPETITIVE PROGRAMMING

Leetcode : Max rating - **2169** | Solved problems: **1250+** | Top **1.06%** globally | Badges : **10** | Highest Rank : **221** | **Guardian**.
Codeforces : Max rating - **1517** | Position: **Specialist** | Solved problems: **300+** | Highest rank **872** in round **1048** division **2**.
Codechef : Max rating - **1869** | Position : **4 Star** | Problems : **133** | Highest rank **158** in Starters 155 (Rated) **167** in division **2**.
Global Rank in Codechef : **5338** | **Guardian** badge in **Leetcode** | Solved **30+** problems in **CSES** | Solved 10+ **Project Euler** .

ACHIEVEMENTS

- Participated in ICPC Chennai Regional 2025, successfully solved a contest problem, and secured an overall rank of 1034.
- Winner of **SIH 2025**, built a geospatial calibration tool for **IKS** to generate precise dimensions for astronomical instruments.
- Built an autonomous drone system using YOLOv5 for QR-based landing , secured AIR 3 in ADDC 2025 drone competition.
- Ranked in Top 10 of Codathon 2024/25, a competitive programming contest organized by my college showcasing my skills.

VOLUNTEERING

- Member of the Japanese Club, actively involved in event management and hosting sessions for visiting Japanese delegates.
- Taught Java programming fundamentals to junior students, delivering hands-on exercises to enhance learning and coding skills
- Coordinated an event during the Trojans of the Technical Symposium, Facilitating seamless implementation of the participants